

Organizational Procurement Management System

System Scope Specification, Architecture Baseline & Delivery Roadmap

Document Ref: OPMS-SPEC-2026-V1	Target Platform: Enterprise Web Application	Commercial Scope: Fixed-Price (₹4,000 INR Allocation)
Production Stack: Laravel 10 / React 18 / MySQL	Prepared By: Pratik Kadole (Devlogic Systems)	Status: Engineering Baseline Approved

1. EXECUTIVE SUMMARY & PROBLEM DEFINITION

Modern domestic enterprises encounter severe operational friction, high administrative overhead, and procurement bottlenecks when relying on manual, decentralized purchasing methods. Traditional procurement cannot satisfy the velocity, auditing standards, and competitive pricing required by modern organizations.

The **Organizational Procurement Management System (OPMS)** provides a centralized digital procurement platform that directly interconnects procuring organizations with verified wholesale suppliers, automating the entire request-for-quotation, competitive bidding, purchase order execution, and fulfillment tracking lifecycle.

2. ARCHITECTURAL MODERNIZATION & TECHNOLOGY SELECTION

Preliminary conceptual synopses made reference to legacy 2010 tooling (Visual Studio 2010, ASP.NET Web Forms, and Microsoft SQL Server 2008). To ensure long-term maintainability, platform security, and immediate deployment viability within the established fixed budget, the solution architecture has been standardized on a modern, open-source enterprise web stack:

Architectural Tier	Selected Production Stack	Technical & Financial Rationale
Application Backend	Laravel 10.x (PHP 8.2 Runtime)	Scalable MVC architecture, Eloquent ORM, integrated CSRF/session protection, zero software licensing fees.
Frontend Presentation	React 18 with Tailwind CSS & Inertia.js	Component-driven single-page responsiveness, zero page flicker, native mobile and tablet adaptive layout.
Database Layer	MySQL 8.0+ / MariaDB Engine	ACID-compliant relational transactions, normalized entity structure, cross-platform host portability.
Authentication & Security	Role-Based Access Control (RBAC) & Bcrypt	Cryptographic hashing, secure HTTP cookies, multi-tenant role isolation (Admin, Buyer, Supplier).

3. IN-SCOPE FUNCTIONAL MODULES (DELIVERED ARCHITECTURE)

The system implements the full operational lifecycle defined in the organizational procurement requirements:

3.1. Authentication & Role-Based Access Control (RBAC)

- Role Segregation:** Distinct access levels and portal dashboards for Procuring Organizations (Buyers), Commercial Sellers (Suppliers), and System Administrators.
- Supplier Verification Workflow:** Two-tier onboarding requiring suppliers to submit commercial registration details and trade licenses for administrative moderation before product catalog activation.
- Platform Administration:** Administrative oversight for vendor verification, user lifecycle management, and catalog compliance.

3.2. Product Catalog & Search Subsystem

- Centralized Marketplace Catalog:** Filterable and searchable product directory categorized by industry groups, product codes, units, and minimum order quantities (MOQ).
- Tiered Volume Wholesale Pricing:** Configurable bulk price schedules granting automated per-unit discounts based on procurement volume thresholds.
- Supplier Inventory Management:** Self-service portal for suppliers to upload product specifications, inventory counts, and commercial terms.

3.3. Request for Quotation (RFQ) & Bidding Engine

- **Formal RFQ Creation:** Procuring organizations draft structured requests specifying item requirements, target quantities, expected delivery schedules, and destination details.
- **Multi-Supplier Bidding:** Verified vendors browse open organizational RFQs and submit itemized competitive quotes.
- **Side-by-Side Quote Comparison:** Evaluation matrix enabling buyers to compare multi-vendor bids across unit costs, delivery timelines, and total value.
- **Quote Award & Conversion:** Single-click acceptance workflow that marks winning quotes and automatically generates associated purchase orders.

3.4. Purchase Order Fulfillment & Invoicing

- **Automated Order Generation:** Direct pipeline transforming accepted bids into formal purchase orders.
- **Lifecycle State Machine:** Order status progression (*Pending* → *Confirmed* → *Processing* → *Shipped* → *Delivered*).
- **Commercial Invoicing:** Standardized invoices detailing line items, applicable taxes, totals, and payment status tracking.

3.5. Negotiation & In-App Messaging

- **Contextual Message Threads:** Direct communication channels bound specifically to active RFQs and orders to audit price negotiations and technical inquiries.

4. SCOPE BOUNDARIES & BUDGET ALLOCATIONS (FIXED SCOPE DEFINITION)

To ensure complete transparency and timely project delivery within the **₹4,000 INR commercial development allocation**, explicit architectural boundaries have been established. Secondary non-core concepts from preliminary literature have been evaluated as follows:

Module / Requirement	Status	Engineering Analysis & Budget Rationale
Core Procurement & RFQ Engine	IN SCOPE	Fully implemented. End-to-end organizational RFQ, multi-vendor quote bidding, purchase ordering, and invoicing.
Modern Web Architecture (PHP/React)	IN SCOPE	Fully implemented. Modernized to eliminate proprietary Microsoft software dependencies and licensing costs.
Buyer, Supplier & Admin Portals	IN SCOPE	Fully implemented. Dedicated dashboards, catalog controls, order history, and direct messaging.
Equipment Leasing / Rental Subsystem	OUT OF SCOPE	Budget Constraint: The concept of dual-mode procurement (Leasing/Rent vs. Purchase) requires an independent financial asset-leasing ERP module with recurring billing cycles, security deposit escrow, and physical return inspection workflows. Within a ₹4,000 budget, engineering is strictly focused on primary direct volume procurement. (<i>Available as an optional Phase 2 enterprise expansion</i>).
Legacy Microsoft Stack (ASP.NET 2010 / MSSQL 2008)	OUT OF SCOPE	Technical Feasibility: 16-year-old deprecated frameworks introduce severe vulnerability risks, lack modern responsive standards, and mandate commercial Windows Server licensing. Replaced with superior open-source equivalents.
Hardware-Level Anti-Shoulder Surfing Visuals	OUT OF SCOPE	Standards Compliance: Replaced theoretical academic UI gimmicks with industry-standard cryptographic Bcrypt hashing and secure HTTP session tokenization.

Contractual Scope Boundary Notice:

Any additions necessitating secondary leasing workflows, third-party payment gateway merchant accounts, or legacy compiler back-porting fall outside the agreed baseline budget and require separate commercial terms and revised delivery schedules.

5. ENGINEERING MILESTONE & DELIVERY ROADMAP

The project development lifecycle follows a structured three-phase release schedule to ensure exhaustive quality assurance and validation prior to final deployment handoff:

Milestone	Target Workstreams	Key Deliverables	State
Milestone 1 Phase 1	System Architecture & Data Modeling	- Relational schema normalization (Users, Suppliers, Products, RFQs, Quotes, Orders) - Multi-role authentication & middleware guards	COMPLETED
Milestone 2 Phase 2	Procurement & Bidding Engine	- Dynamic product catalog with volume pricing breaks - RFQ submission, supplier quote submission & comparison matrix - In-app contextual negotiation messaging	IN PROGRESS
Milestone 3 Phase 3 (Final)	Order Execution, Verification & Handoff	- Order status lifecycle machine & automated invoice generation - Seed data generation across all user personas - Production asset bundling, documentation & deployment package	SCHEDULED

6. DELIVERY PACKAGE & EVALUATION GUIDELINES

The final release package delivered upon project sign-off contains:

- 1. Complete Verified Source Code:** Clean, unencumbered application repository containing all backend models, controllers, migrations, and React/Tailwind frontend components.
- 2. Automated Database Seeder:** Pre-populated evaluation environment including Admin, Corporate Buyer, and Verified Commercial Supplier accounts.
- 3. Technical Documentation:** Comprehensive configuration guide with step-by-step local server bootstrap instructions.

7. FORMAL ACCEPTANCE & SIGN-OFF

Delivery is deemed satisfied upon provision of the packaged source repository and verified demonstration of the core procurement, RFQ, and bidding workflows outlined in Section 3.

Prepared By: Pratik Kadole Engineering Lead / Solutions Delivery Devlogic Systems Date: September 2026	Acknowledged & Accepted By: Procurement Review Lead / Client Representative Project Reference: OPMS-SPEC-2026-V1 Date: _____
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